Snowflake Architecture

By
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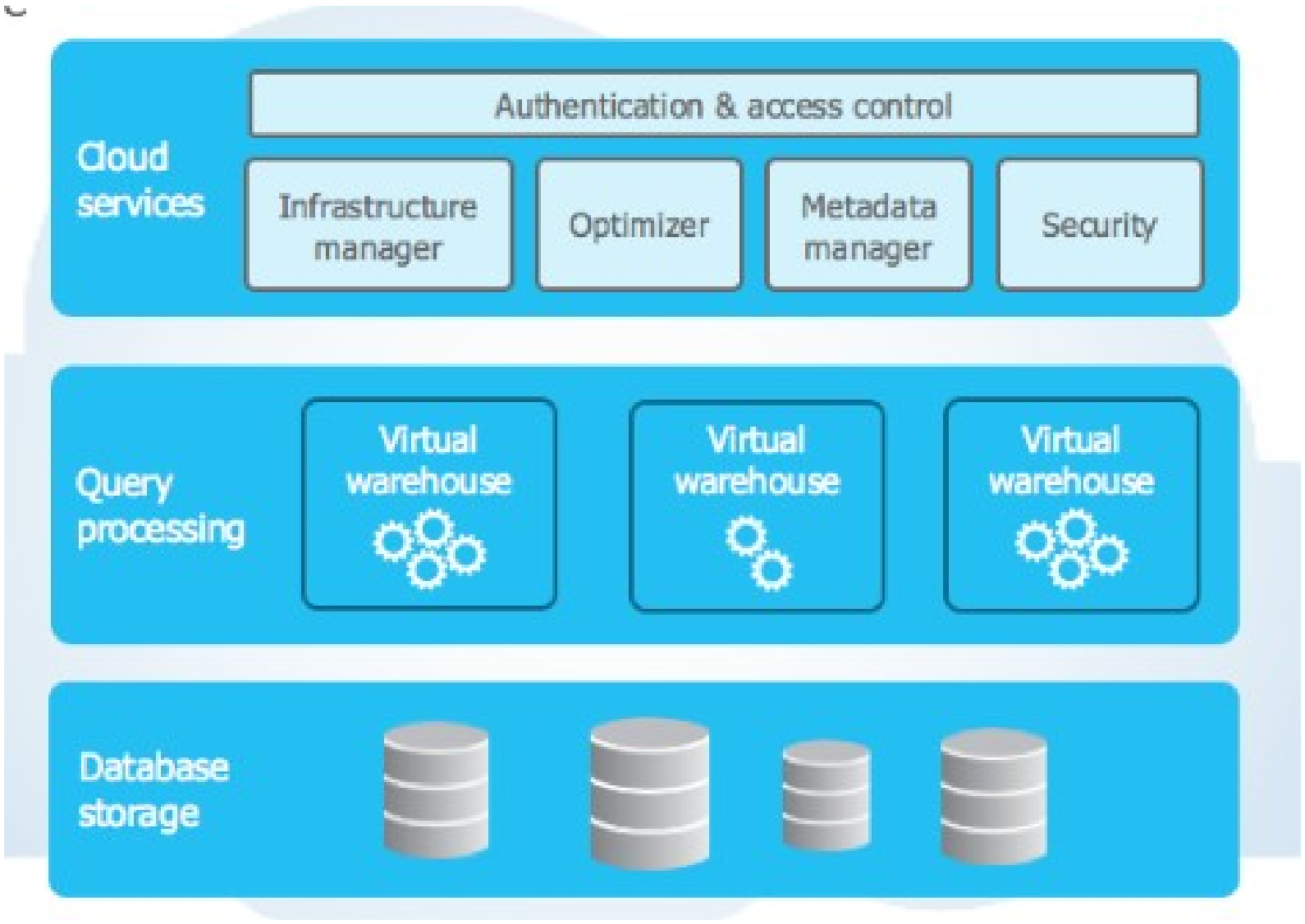

Cloud services
Authentication & access control
Infrastructure manager
Optimizer
Metadata manager
Security
Query processing
Virtual warehouse
Virtual warehouse
Virtual warehouse
Database storage

Database Storage Layer:

1. Stores table data and query results
2. Data will be stored in columnar format
3. Data will be stored in micro partitions
4. Snowflake manages all aspects of how this data is stored i.e. the data organization, file size, structure, compression, metadata, statistics
5. The data objects stored by Snowflake are not directly visible nor accessible by customers; they are only accessible through SQL query operations run using Snowflake.
6. We can define cluster keys on large tables for better performance.

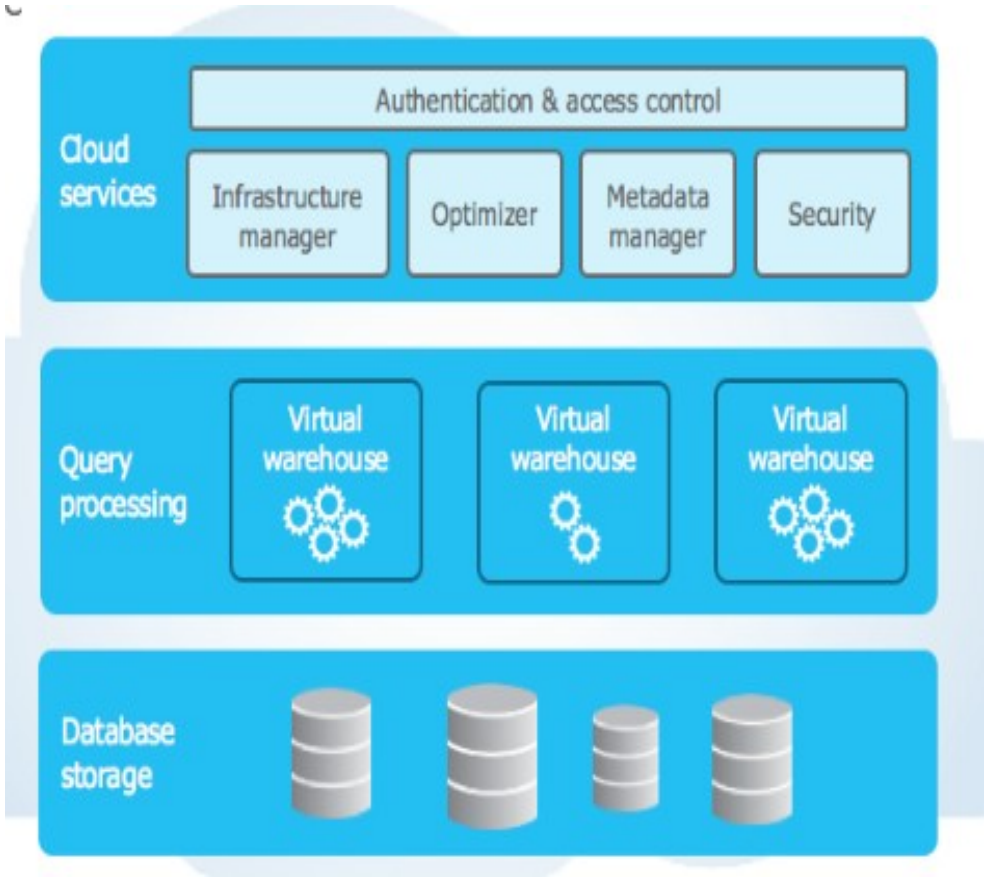
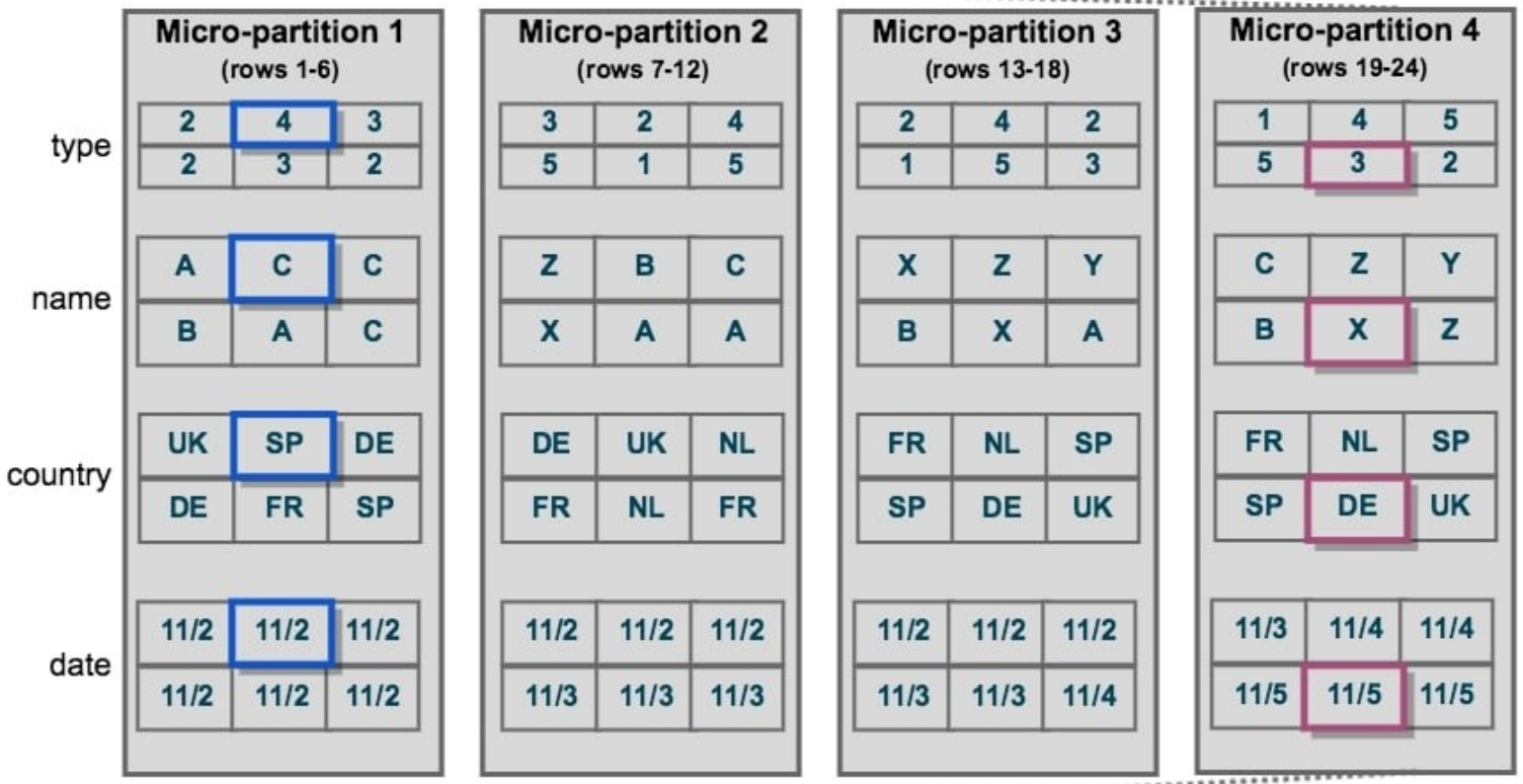


Table: `t1`

Logical Structure

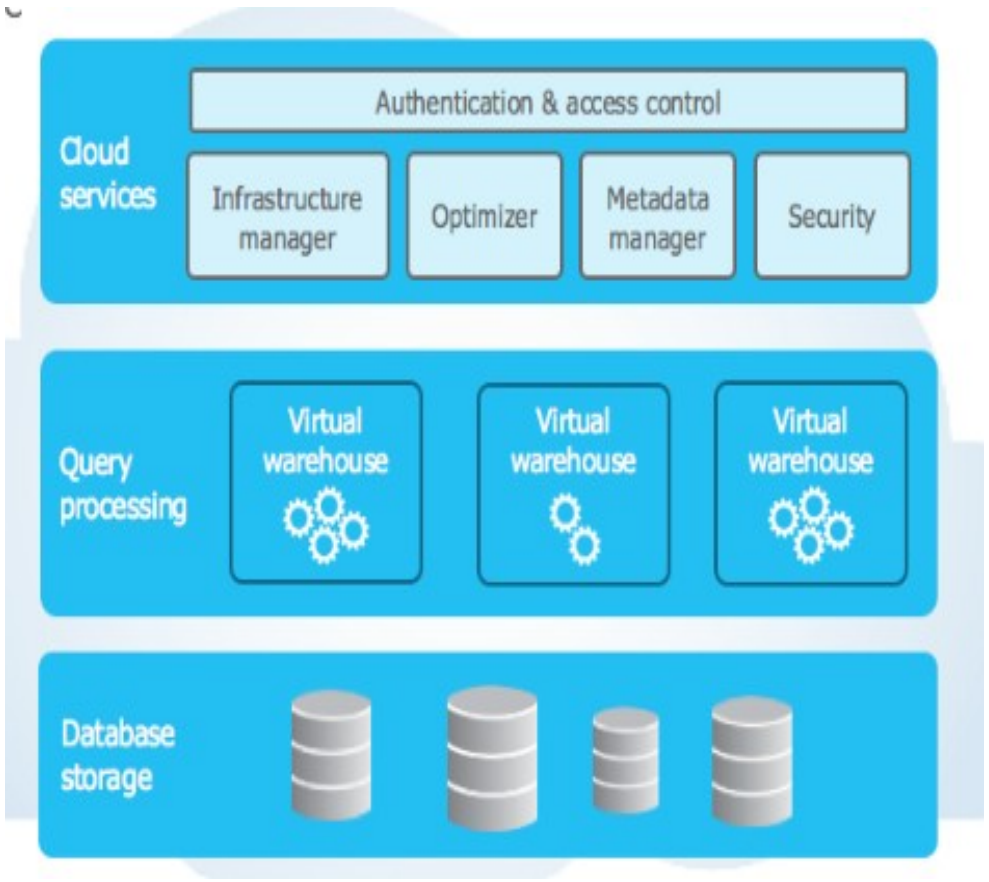
type	name	country	date
2	A	UK	11/2
4	C	SP	11/2
3	C	DE	11/2
2	B	DE	11/2
3	A	FR	11/2
2	C	SP	11/2
3	Z	DE	11/2
2	B	UK	11/2
4	C	NL	11/2
5	X	FR	11/3
1	A	NL	11/3
5	A	FR	11/3
2	X	FR	11/2
4	Z	NL	11/2
2	Y	SP	11/2
1	B	SP	11/3
5	X	DE	11/3
3	A	UK	11/4
1	C	FR	11/3
4	Z	NL	11/4
5	Y	SP	11/4
5	B	SP	11/5
3	X	DE	11/5
2	Z	UK	11/5

Physical Structure



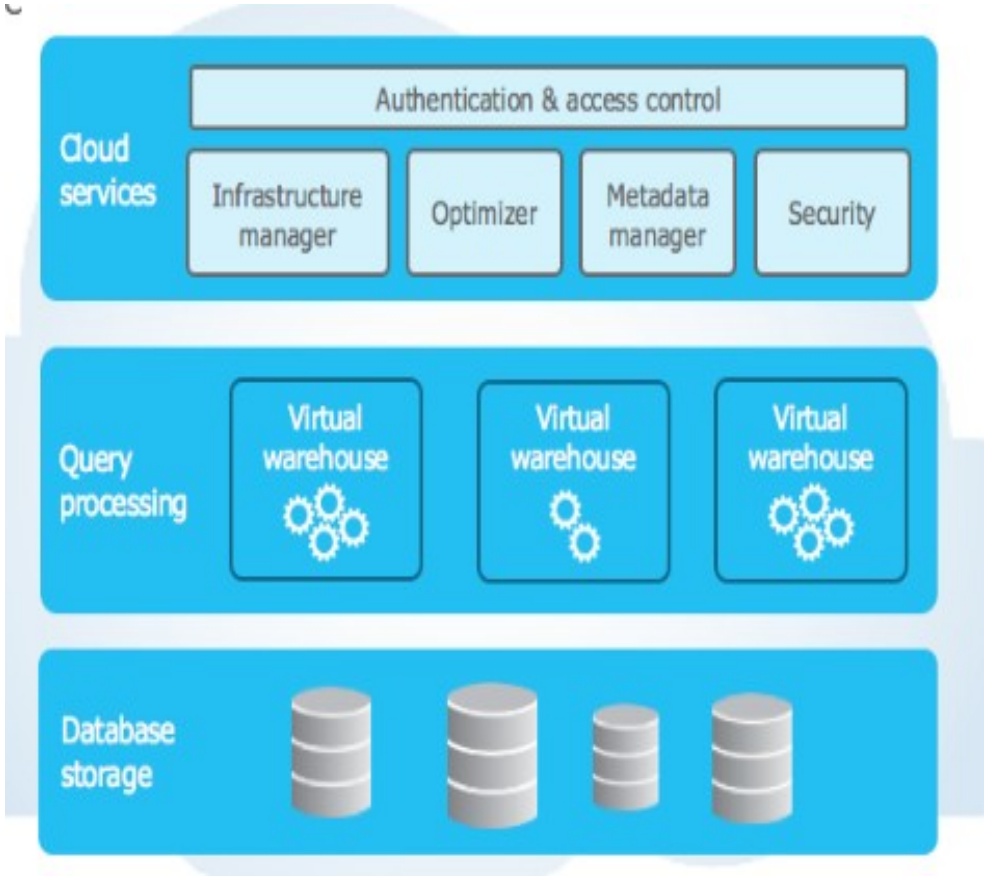
Query Processing Layer:

1. This is the actual processing unit of snowflake
2. Snowflake processes queries using “virtual warehouses”
3. Each virtual warehouse is composed of multiple compute nodes allocated by Snowflake from a cloud provider.
4. On AWS they are a group of EC2 instances and on AZURE a group of Virtual Machines
5. Compute cost will be calculated on the basis of query execution time on virtual warehouses
6. Virtual Warehouses are considered as the muscle of the system
7. Can scale up and scale down easily
8. Auto-Suspend and Auto-Resume is available



Cloud Services Layer:

1. Collection of services that coordinate activities across Snowflake
2. This is the Brain of the snowflake
3. Authentication and access control
4. Infrastructure management
5. Metadata Management
6. Security
7. Manages all Serverless tasks like Snowpipe, Tasks, Materialized view maintenance etc.



Connecting to Snowflake

Snowflake supports multiple ways of connecting to the service:

- A web-based user interface from which all aspects of managing and using Snowflake can be accessed.
- Command line clients (e.g. SnowSQL) which can also access all aspects of managing and using Snowflake.
- ODBC and JDBC drivers that can be used by other applications (e.g. Tableau) to connect to Snowflake.
- Through native connectors available in ETL tools (e.g. Datastage, Informatica)

Thank you